

4 A2 - Temă seminar 7

1. $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$$T(x, y, z) = (x + 2y, 2y, -2y + z)$$

T apl liniară (x matriceală)

$$T(x) = Ax, \text{ unde } x = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \text{ unde } A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ 0 & -2 & 1 \end{pmatrix} \in M_3(\mathbb{R})$$

fie $x_1, x_2 \in \mathbb{R}^3$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \quad \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$$

unde $\alpha_1, \alpha_2 \in \mathbb{R}$

$$\begin{aligned} T(\alpha_1 x_1 + \alpha_2 x_2) &= A(\alpha_1 x_1 + \alpha_2 x_2) = A(\alpha_1 x_1) + A(\alpha_2 x_2) = \\ &= (A \alpha_1) x_1 + (A \alpha_2) x_2 = (\alpha_1 A) x_1 + (\alpha_2 A) x_2 = \alpha_1 (A x_1) + \\ &+ \alpha_2 (A x_2) = \alpha_1 T(x_1) + \alpha_2 T(x_2) \Rightarrow T \text{ apl lin} \end{aligned}$$

2. $\mathcal{A}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$$\mathcal{A}(x, y, z) = (2x - y + 2z, -x + 2y - z, x + y + z)$$

a) $A_{\mathcal{A}} = \begin{pmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{pmatrix} \in M_3(\mathbb{R})$

$\begin{matrix} \text{unde} & \text{unde} & \text{unde} \\ \mathcal{A}(e_1) & \mathcal{A}(e_2) & \mathcal{A}(e_3) \end{matrix}$

b) $P(\lambda) = \det(A_{\mathcal{A}} - \lambda I_3) = \begin{vmatrix} 2-\lambda & -1 & 2 \\ -1 & 2-\lambda & -1 \\ 1 & 1 & 1-\lambda \end{vmatrix} =$

$$= -\lambda(\lambda - 2)(\lambda - 3)$$

$$p(\lambda) = 0 \Leftrightarrow -\lambda(\lambda-2)(\lambda-3) = 0 \Rightarrow$$

$$\Rightarrow \begin{cases} \lambda_1 = 0 \\ \lambda_2 = 2 \in \mathbb{R} \\ \lambda_3 = 3 \end{cases} \quad \text{Spec}(A) = \{0, 2, 3\}$$

$$m_0(\lambda_1) = m_2(\lambda_2) = m_3(\lambda_3) = 1$$

Subspace problem:

$$S_{\lambda_1=0} \begin{cases} 2x - y + 2z = 0 \\ -x + 2y - z = 0 \\ x + y + z = 0 \end{cases} \Rightarrow \text{is lin omogen} \\ \text{3 ec, 3 rec}$$

$$\text{rang}(A - \lambda_1 I_3) = 0$$

$$A - \lambda_1 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 3 \neq 0 \Rightarrow \begin{cases} x, y \text{ rec wrt} \\ z = \alpha \text{ rec rec} \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} 2x - y = -2x \\ -x + 2y = \alpha \end{cases} \Rightarrow \begin{cases} x = -x \\ y = 0 \end{cases}$$

$$\text{deci } V_{\lambda_1} = \{(-x, 0, x) \mid x \in \mathbb{R}\} = \{x(-1, 0, 1) \mid x \in \mathbb{R}\} \\ \stackrel{n}{V_1}$$

$$V_{\lambda_1} = \mathbb{R} \cdot v_1$$

$$S_{\lambda_2=2} \begin{cases} 2x - y + 2z = 2x \\ -x + 2y - z = 2y \\ x + y + z = 2z \end{cases} \Rightarrow \begin{cases} -y + 2z = 0 \\ -x - z = 0 \\ x - y + z = 0 \end{cases}$$

$$A - \lambda_2 = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = 1 \neq 0 \Rightarrow \text{rang}(A - \lambda_2 I_3) = 2$$

$$\Rightarrow \begin{cases} x, y \text{ free} \\ z = \mu \text{ free} \end{cases}$$

$$\Rightarrow \begin{cases} z = 2\mu \\ x = -\mu \end{cases} \Rightarrow$$

$$\text{defi } v_{x2} = \underbrace{\mu(-1, 2, 1)}_{v_2} \mid \mu \in \mathbb{R}$$

$$v_{x2} = \mu(v_2)$$

$$S_{\lambda_3=3} \begin{cases} -x - y + 2z = 0 \\ -x - y - z = 0 \\ x + y - 2z = 0 \end{cases}$$

$$\Delta_p = \begin{vmatrix} -1 & 2 \\ -1 & -1 \end{vmatrix} = 3 \neq 0 \Rightarrow \text{rg}(A - \lambda_3 I_3) = 2$$

$$\Rightarrow \begin{cases} x, z \text{ free} \\ y = \tau \text{ free} \end{cases}$$

$$\begin{cases} -x + 2z = \tau \\ -x - z = \tau \end{cases} \Rightarrow \begin{cases} z = 0 \\ x = -\tau \end{cases} \Rightarrow$$

$$\text{defi } v_{x3} = \underbrace{\tau(-1, 1, 0)}_{v_3} \mid \tau \in \mathbb{R}$$

$$v_{x3} = \tau(v_3)$$

c) stud de este diag:

$$m_a(x_1) = m_a(x_2) = m_a(x_3) = 1$$

$$\left. \begin{aligned} v_{\lambda_1} &= \eta \langle v_1 \rangle \Rightarrow \dim v_{\lambda_1} = 1 = m_a(x_1) \\ v_{\lambda_2} &= \eta \langle v_2 \rangle \Rightarrow \dim v_{\lambda_2} = 1 = m_a(x_2) \\ v_{\lambda_3} &= \eta \langle v_3 \rangle \Rightarrow \dim v_{\lambda_3} = 1 = m_a(x_3) \end{aligned} \right\} \quad \Rightarrow$$

matr a este diag

$$d) \quad D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$B = \{ v_1 = (-1, 0, 1), v_2 = (-1, 2, 1), v_3 = (-1, 1, 0) \}$$